4
Fibonacci Numbers

4.1 Fibonacci's Exercise

In the thirteenth century, the Italian mathematician Leonardo Fibonacci studied the following (not too realistic) question:



Leonardo Fibonacci

A farmer raises rabbits. Each rabbit gives birth to one rabbit when it turns 2 months old, and then to one rabbit each month thereafter. Rabbits never die, and we ignore male rabbits. How many rabbits will the farmer have in the nth month if he starts with one newborn rabbit?

It is easy to figure out the answer for small values of n. The farmer has 1 rabbit in the first month and 1 rabbit in the second month, since the rabbit has to be 2 months old before starting to reproduce. He has 2 rabbits during the third month, and 3 rabbits during the fourth, since his first rabbit delivered a new one after the second and one after the third. After 4 months, the second rabbit also delivers a new rabbit, so two new

rabbits are added. This means that the farmer will have 5 rabbits during the fifth month.

It is easy to follow the multiplication of rabbits for any number of months if we notice that the number of new rabbits added each months is just the same as the number of rabbits who are at least 2 months old, i.e., who were already there in the previous month. In other words, to get the number of rabbits in the *next* month, we have to add the number of rabbits in the *previous* month to the number of rabbits in the *current* month. This makes it easy to compute the numbers one by one:

$$1,\ 1,\ 1+1=2,\ 2+1=3,\ 3+2=5,\ 5+3=8,\ 8+5=13,\ \ldots$$

(It is quite likely that Fibonacci did not get his question as a real applied math problem; he played with numbers, noticed that this procedure gives numbers that were new to him but nevertheless had very interesting properties—as we'll see ourselves—and then tried to think of an "application.")

To write this as a formula, let us denote by F_n the number of rabbits during the nth month. Then we have, for $n = 2, 3, 4, \ldots$,

$$F_{n+1} = F_n + F_{n-1}. \tag{4.1}$$

We also know that $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$. It is convenient to define $F_0 = 0$; then equation (4.1) will remain valid for $n = 1$ as well. Using equation (4.1), we can easily determine any number of terms in this sequence of numbers:

$$0,\ 1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ 55,\ 89,\ 144,\ 233,\ 377,\ 610,\ 987,\ 1597, \ldots .$$

The numbers in this sequence are called *Fibonacci numbers*.

We see that equation (4.1), together with the special values $F_0 = 0$ and $F_1 = 1$, uniquely determines the Fibonacci numbers. Thus we can consider (4.1), together with $F_0 = 0$ and $F_1 = 1$, as the definition of these numbers. This may seem a somewhat unusual definition: Instead of telling what F_n is (say, by a formula), we just give a rule that computes each Fibonacci number from the two previous numbers, and specify the first two values. Such a definition is called a *recurrence*. It is quite similar to induction in spirit (except that it is not a proof technique, but a definition method), and is sometimes also called *definition by induction*.

4.1.1 Why do we have to specify exactly two of the elements to begin with? Why not one or three?

Before trying to say more about these numbers, let us consider another counting problem:

> *A staircase has n steps. You walk up taking one or two at a time. How many ways can you go up?*

For $n = 1$, there is only 1 way. For $n = 2$, you have 2 choices: take one step twice or two once. For $n = 3$, you have 3 choices: three single steps, or one single followed by one double, or one double followed by one single.

Now stop and try to guess what the answer is in general! If you guessed that the number of ways to go up on a stair with n steps is n, you are wrong. The next case, $n = 4$, gives 5 possibilities ($1+1+1+1$, $2+1+1$, $1+2+1$, $1+1+2$, $2+2$).

So instead of guessing, let's try the following strategy. Let's denote by J_n the answer. We try to figure out what J_{n+1} is, assuming we know the value of J_k for $1 \leq k \leq n$. If we start with a single step, we have J_n ways to go up the remaining n steps. If we start with a double step, we have J_{n-1} ways to go up the remaining $n-1$ steps. These are all the possibilities, and so

$$J_{n+1} = J_n + J_{n-1}.$$

This equation is the same as the equation we have used to compute the Fibonacci numbers F_n. Does this means that $F_n = J_n$? Of course not, as we see by looking at the beginning values: for example, $F_3 = 2$ but $J_3 = 3$. However, it is easy to observe that all that happens is that the J_n are shifted by one:

$$J_n = F_{n+1}.$$

This is valid for $n = 1, 2$, and then of course it is valid for every n, since the sequences $F_2, F_3, F_4, \ldots$ and $J_1, J_2, J_3, \ldots$ are computed by the same rule from their first two elements.

4.1.2 We have n dollars to spend. Every day we buy either a candy for 1 dollar or an ice cream for 2 dollars. In how many ways can we spend the money?

4.1.3 How many subsets does the set $\{1, 2, \ldots, n\}$ have that contain no two consecutive integers?